Data Visualisation for Social Scientists - Final Project

Ellen Beattie

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Executive Summary

This project aims to investigate how the United Kingdom measures up in the development and integration of Artificial Intelligence (AI). Through the use of visualisations I will explore how the UK compares to other countries on the world stage, specifically in terms of economic development, public opinion and ethical/responsible practices and governance. By visualising the data I aim to gain key insights into the topic and capture a snapshot of the current state of the UK with regards to AI.

Data Background

The data used in this project is from the a Global Artificial Intelligence Indicator Database (GAID) 1998-2025, sourced from the Harvard Dataverse. This longitudinal panel dataset sources, standardises and consolidates indicator information from Stanford's AI Index, OECD.ai and the Global Index on Responsible AI (GIRAI). It contains information from 214 countries, across 28 years with data on 24,323 unique metrics. There is varying levels of coverage across the data, and the metrics are grouped into key source categories such as Economy, Education, Public Opinion etc. There is a comprehensive codebook accessible with detailed descriptions of the topics, specific metrics and the units of measurement.

Data Cleaning

After downloading and importing the CSV file, I explored the underlying data structure. The data was provided in long form, making it easier to filter and clean for plotting each individual figure. Packages were loaded along with the data.

```r
lapply(c("dplyr","tidyverse", "ggplot2", "readr", "readxl", "showtext", "purrr",
         "ggridges", "ggdist", "xtable", "ggrepel", "extrafont", "WDI", "ggtext",
         "ggdist", "forecats", "patchwork",  "sf", "WDI", "regions", "tidytext",
         "gutenbergr"),  pkgTest)
loadfonts(device = "win")
```

```r
#### Load data ####
df <- read.csv("MASTER_AI_DATA_COMPILATION_FINAL.csv")
uk <- "United Kingdom"
```

For simplicity and consistency I created a base theme to apply to all the figures, as well as a base colour palette.

```r
#### Creating Base Theme ####
theme_ellen <- function(...) {
  theme_classic(base_size = 10, base_family = "CMU Sans Serif") +
    theme(
      # Titles
      plot.title = element_text(face = "bold", size = rel(1.3)),
      plot.subtitle = element_text(face = "plain", size = 9, color = "grey50"),
      plot.caption = element_text(face = "italic", size = rel(0.7),
                                  color = "grey60", hjust = 0),
      #Backgrounds
      # Grid
      panel.background = element_rect(fill = "white", colour = NA), #clean base to work from
      panel.border = element_blank(),
      panel.grid = element_blank(),
      panel.grid.minor = element_blank(),

      #Axis
      axis.title = element_text(family = "CMU Sans Serif", size = 10), #title font
      axis.line.x = element_line(linewidth = 0.4), #axis lines tin
      axis.line.y = element_line(linewidth = 0.4),
      axis.text = element_text(colour = "black", size = 8), #axis text change
      axis.ticks.x = element_line(colour = "black", linewidth = 0.75),

      #Facets
      strip.background = element_blank(),
      strip.text = element_text(size = 11),

```

```
      #Legend
      legend.title = element_text(family = 'CMU Sans Serif', size = 10, colour = 'black'), #legend
      legend.text =  element_text(family = 'CMU Sans Serif', size = 8, colour = 'black'),# font change
      legend.key.height = unit(1, "lines"),#spacing
      legend.key.width = unit(1, "lines"),
      legend.spacing.y = unit(1, "lines"),
      legend.background = element_rect(
        fill = NA,
        colour = "grey60", #make box around legend to align it
        size = 0.5,
        linetype = "solid")
    )
}
```

Individual Figures

0.1 Figure 1: Plot of AI Vibrancy Scores Economy vs R&D

First, I wanted to understand the landscape of global AI development. To do this I created Figure 1 which plots the relative positions of major world countries on two axes: Research and Development describing the countries' AI innovation capacity including components such as AI journal/conference publications and AI-related patent fillings/grants; and Economic describing the countries' capacity to exploit AI including talent intensity, investment, new AI companies and AI skill penetration. The normalised vibrancy score on these two dimension is sourced from the Stanford Global Vibrancy index.

```
#### Figure 1: AI Vibrancy Economy vs R&D
AI_vib_scores_wide <- df %>%
  filter(Source_Category == "Global AI Vibrancy Tool",
         Year == 2021, #latest year updated
         Metric %in% c(
           "AI Vibrancy Index Pillar Score: Economy (0-100)",
           "AI Vibrancy Index Pillar Score: Research and Development (0-100)") %>%
  select(Country, Metric, Value) %>% #rid excess columns
  pivot_wider(
    names_from = Metric,
    values_from = Value) %>%
  rename(
    `Economy` = `AI Vibrancy Index Pillar Score: Economy (0-100)`,
    `R&D` = `AI Vibrancy Index Pillar Score: Research and Development (0-100)`) %>%
  mutate(
    Highlight = ifelse(Country == "United Kingdom", "UK", "Other")) #for colour distinction

```

```r
figure1 <- ggplot(AI_vib_scores_wide, aes(x = Economy, y = `R&D`, color = Highlight)) +
  geom_point(size = 1.5, alpha = 1) + #larger points
  geom_text_repel(aes(label = Country), size = 3, show.legend = FALSE, #text removing legend
                  family = 'CMU Sans Serif', segment.color = NA) + #specifying same font
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey90") + # 45 degree line
  scale_x_continuous(limits = c(0, 80),expand = c(0,0)) + #to see cluster more clearly
  scale_y_continuous(limits = c(0, 80),expand = c(0,0)) +
  scale_color_manual(values = c("UK" = "#fd3838", "Other" = "#4b6cf4")) + #differentiate UK
  guides(colour = "none") + #remove highlight legend
  theme_ellen()+ #add theme
  labs(
    title = "AI Vibrancy: Economy vs Research and Development by Country",
    subtitle = "Economy and R&D pillar scores for a given country on a 0-100 scale \n measure the normalised aggregate performance of AI vibrancy metrics",
    x = "Economy Score",
    y = "R&D Score",
    caption = "Source: Stanford AI Index - Global AI Vibrancy Tool (2021). \n Economic score components include hiring intensity, newly funded companies, investment, skill penetration. \n R&D score components include the number of AI journal/conference/public repository publications and citations and AI-related patents fillings/grants")

figure1
ggsave(figure1, filename = "figures/figure1.pdf", device = cairo_pdf,
       width = 6, height = 4, units = "in", bg = "transparent")
```

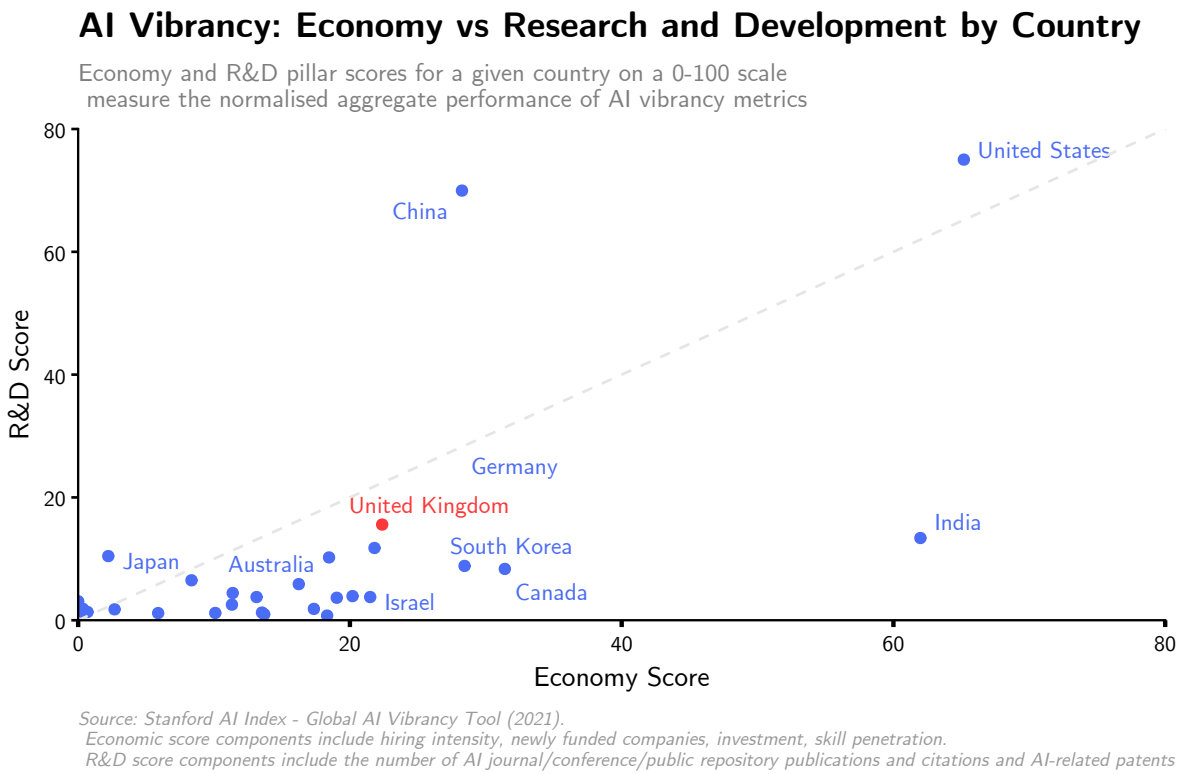


Figure 1 shows that the US, China and India are clear outliers in different ways. The US has a leadership position in both dimensions, whereas China appears to be less able to capitalise economically on its strong capability in AI R&D. Conversely, India is well-positioned to economically exploit AI but is less productive in terms of innovation measures. Within the chasing pack the UK ranks third in R&D but sixth in economic development. Generally most other countries appear to perform better on economic metrics than on R&D.

0.2 Figure 2: Bar Chart of Total Private Investment in AI

To better understand the level of private investment in AI, I analysed the level of private capital investment per country over the last twelve years. Figure 2 shows a bar chart of the cumulative amount of investment (2013-2024) of the top countries in the world, highlighting the UK by colour and adding the value (in billions USD) to each bar to easily display the total private capital invested in each country.

```r
#### Figure 2: Cumulative Private Investment
cumulative_priv_investment <- df %>%
  filter(Source_Category == "Economy") %>%
  filter(Metric %in% c("Total Private Investment in AI (In Billions Of US
    Dollars) [Cumulative 2013-2024]")) %>%
  select(Country, Value) %>%
  filter(Value > 3) %>% #just focus on top economies
```

```r
  group_by(Country) %>%
  ungroup() %>%
  mutate(Country = forcats::fct_reorder(Country, Value), #reorder by size of metric
         Highlight = ifelse(Country == uk, "Highlighted", "Other")) #add highlight column


figure2 <- ggplot(data = cumulative_priv_investment, aes(x = Country, y = Value, fill = Highlight)) +
  geom_col() +
  geom_text(aes(label = round(Value,1)), vjust = -0.3, size = 2) + #add text of value to top
  scale_fill_manual(values = c("Highlighted" = "#a72264", "Other" = "#b6c3e8") ) + #highlight uk
  scale_y_continuous(limits = c(0, 550), expand = c(0,0)) +
  theme_ellen() +
  theme(
    axis.text.x = element_text(hjust = 1, angle = 30, size = 8),
    axis.text.y = element_text(size = 8),
    legend.position = "none",
    axis.title.x = element_blank(),
  ) +
  labs(
       y = "Billions USD",
       title = "Total Private Investment in AI-related Companies 2013-2024",
       subtitle = "Cumulative private capital invested in a given country over twelve years",
       caption = "Source: Stanford AI Index - Economy 2025. Measures the total aggregate amount of private capital (in billions of US dollars) invested
in AI-related companies in a given country over the twelve-year period")

figure2
ggsave(figure2, filename = "figures/figure2.pdf", device = cairo_pdf,
       width = 6, height = 4, units = "in", bg = "transparent")
```

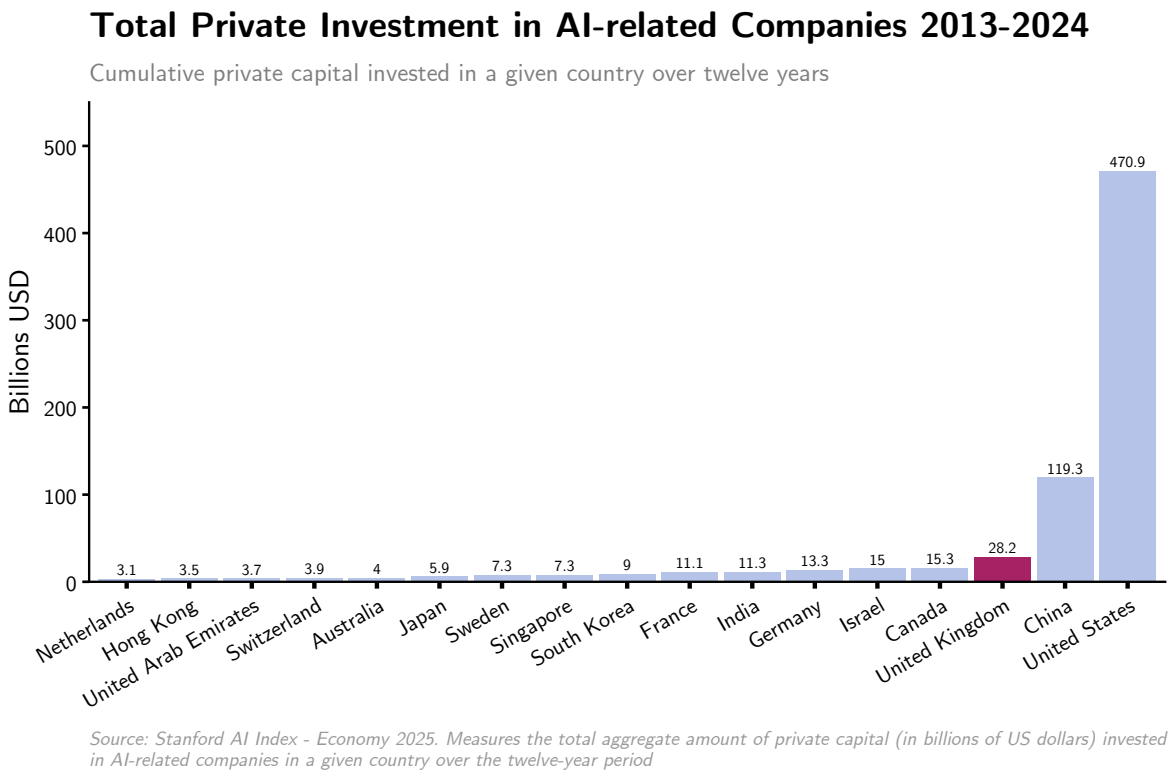


Figure 2 shows how much more the US (and to some degree China) has invested privately in AI compared to the rest of the world. The UK was surprisingly high relative to other comparable countries, coming in third globally with 28.2 billion dollars of investment being recorded between 2013 and 2024. This suggests that people have determined that AI companies in the UK are worth investing in and may be profitable, managing to gain more investment than any other European nations of similar size.

0.3 Figure 3: Percentage Stacked Bar Chart of Female AI Talent Representation

I wanted to look further into the UK's AI capabilities, and in particular at the workforce composition and whether AI talent is spread across genders. To do this I plotted a percentage stacked bar chart to easily visualise what proportion of AI specialists identified as female versus male. By lining up the countries it is easy to determine how much different workforces differ in terms of gender representation and how the UK compares.

```
#### Figure 3: Global Talent Representation Gender in 2024
AI_gender_2024 <- df %>%
  filter(Metric %in% c("AI Talent Representation: Female",
                       "AI Talent Representation: Male"),
         Year == 2024,
```

```r
        !Country %in% c("Czechia", "Cyprus", "Luxembourg", "Costa Rica", "Estonia", "Lithuania", "Israel", "Greece", "Croatia", "Poland", "Denmark", "Turkey","New Zealand")) %>% #remove some countries to have more focus on uk (even split above/below)
  select(Country, Metric, Value) %>%
  pivot_wider(names_from = Metric, values_from = Value) %>%
  rename(Female = `AI Talent Representation: Female`,
         Male = `AI Talent Representation: Male`) %>%
  arrange(Female) %>% #arrange by order
  mutate(Country = factor(Country, levels = Country)) %>%
  pivot_longer(cols = c(Female, Male), names_to = "Gender", values_to = "Percentage") %>% #so each country factor for both
  mutate(
    Highlight = ifelse(Country == "United Kingdom", "Highlighted", "Other"),
    Fill = paste(Gender, Highlight, sep = "."), #create 4 categories of colours for each combo
    Fill = factor(Fill, levels = c("Male.Other","Male.Highlighted","Female.Other","Female.Highlighted"))
  )


figure3 <- ggplot(AI_gender_2024, aes(x = Country, y = Percentage, fill = Fill)) +
  geom_bar(stat = "identity", position = "fill", linewidth = 0.75, color = "white") + #bars
  coord_flip() +
  geom_text(data = AI_gender_2024,
            aes(label = paste0(round(Percentage,1),"%")), #add value in middle
            position = position_fill(vjust = 0.5), #exact center
            color = "white", size = 1.5, fontface = "bold") + #in white to stand out
  scale_y_continuous(labels = scales::percent_format(), #as percentage
                     expand = c(0,0)) +
  scale_fill_manual(values = c(
    "Female.Highlighted" = "#a72264", #colour for 4 options of labels
    "Female.Other" = "#ed90af",
    "Male.Highlighted" = "#5e72c5",
    "Male.Other" = "skyblue"
  )) +
  theme_ellen() +
  theme(legend.position = "none", #remove colour option
        axis.title.y = element_blank(),
        axis.text.y = element_text(size = 6),
        axis.title.x = element_text(size = 8),
        axis.text.x = element_text(size = 8)) +
  labs(title = "2024 AI Talent Representation by Gender",
       subtitle = "Percentage of AI Specialists Identifying as Female versus Male",
       caption = "Source: Stanford AI Index - Economy, \n Measures the share of the total AI workforce in a given country that identifies as female, indicating gender diversity within the AI talent pool",
```

```r
        x = NULL, y = "Percentage %")

figure3
ggsave(figure3, filename = "figures/figure3.pdf", device = cairo_pdf,
       width = 6, height = 4, units = "in", bg = "transparent")
```

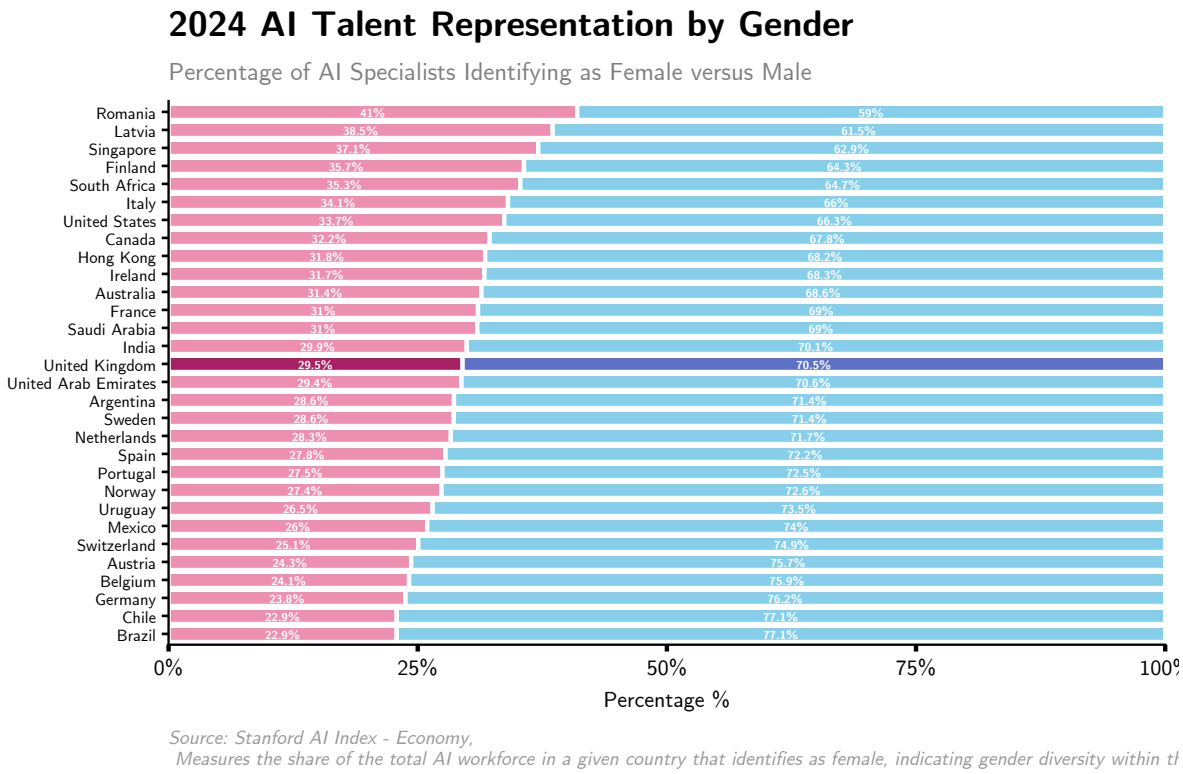


Figure 3 shows that, globally, women are under-represented within the AI-talent workforce. The UK in 2024 appeared solidly in the middle of the pack in terms of female talent representation with 29.5% female to 70.5% male. The US is higher by about 5% whereas Germany, Chile and Brazil have the lowest representation with less than a quarter of their AI talent being female. Interestingly there does not appear to be a clear trend in nation location, investment and gender equality, but it is concerning to see the gender disparity in a sector that is projected to have massive implications for the economy. For the UK specifically, given the high level of overall investment, the country might consider what advantage it could gain by investing in female AI talent.

0.4 Figure 4: Bar Chart of Public Opinion Impact of AI on Daily Life in the Past and Future

Moving beyond these economic and R&D factors, I was also interested in understanding how the UK public think and feel about AI. Firstly I created a bar chart using the Stanford

Public Opinion index based on survey data that describes whether people feel that AI has profoundly changed their life in the past 3-5 years and their projections about whether it will in the next 3-5 years. As the metric from the survey refers to the percentage that agree with a specific statement, and as in all cases there was a higher percentage agreeing on future impact, a dodged bar chart was used. This allows both percentages to be plotted together for each country allowing for easy comparison of each between countries as well as the size of the difference between the two metrics.

```r
#### Figure 4:  Public Opinion - Past and Future impact of AI
AI_change_life <- df %>% filter(Source_Category == "Public Opinion") %>%
  filter(Year == 2025) %>% #latest year
  filter(Metric %in% c("% Agreeing With Statement: Products And Services Using Artificial Intelligence Have Profoundly Changed My Daily Life In The Past 3-5 Years",
                       "% Agreeing With Statement: Products And Services Using Artificial Intelligence Will Profoundly Change My Daily Life In The Next 3-5 Years")) %>%
  mutate(Metric = recode(Metric, #simplify statments
                         "% Agreeing With Statement: Products And Services Using Artificial Intelligence Have Profoundly Changed My Daily Life In The Past 3-5 Years" = "Profoundly Changed Daily Life in Past 3-5 Years",
                         "% Agreeing With Statement: Products And Services Using Artificial Intelligence Will Profoundly Change My Daily Life In The Next 3-5 Years" = "Profoundly Changed Daily Life in Next 3-5 Years"),
         Year = as_factor(Year)) %>%
  group_by(Metric) %>%                    # handle each facet separately
  arrange(Value) %>%                      # sort by value to in order
  mutate(Country = factor(Country, levels = unique(Country))) %>%  # set factor levels
  ungroup() %>%
  mutate(Highlight = ifelse(Country == "United Kingdom", "UK", "Other")) # highlight uk


metric_colours <- c(
  "Profoundly Changed Daily Life in Past 3-5 Years" = "#19765e",
  "Profoundly Changed Daily Life in Next 3-5 Years" = "#94f5dc"
)

figure4 <- ggplot(AI_change_life, aes(x = Country, y = Value, fill = Metric)) +
  geom_col(width = 0.6, alpha = 0.7, position = "identity") + #columns for both in place
  coord_flip() + #horizontal
  annotate("text", #add text annotation
           x = "United Kingdom", y = 62,
           label = "58% of UK Agree AI Will \nProfoundly Change their Life \nin the Next 3-5 Years",
           hjust = 0, size = 2.5,
           color = "#a72264", fontface = "bold") +
  annotate("curve",
```

```
            x = "United Kingdom", xend = "United Kingdom", y = 62, yend = 59,
            arrow = arrow(length = unit(0.15, "cm")),
            curvature = 0.1, color = "#a72264", linewidth = 0.7) +
  scale_y_continuous(expand = c(0,1)) +
  scale_fill_manual(values  = metric_colours) + #pair metric colours selected
  guides(fill = "none")+ #remove fill legend
  theme_ellen() +
  theme(
    axis.text.x = element_text(size = 8),
    axis.text.y = element_text(size = 8),
    plot.subtitle = element_markdown(size = 8, face = "bold", colour = "black") #away from theme
    ) +
  labs( #rather than legend use subtile colour to show what each colour means
    x = NULL, y = "% Agreeing",
    subtitle = paste0(
      "% Agreeing with Statement that AI has Profoundly Changed their Daily Life in the<br> ",
      "<span style='color:", metric_colours["Profoundly Changed Daily Life in Past 3-5 Years"], "'>Past 3-5 Years</span> ",
      " and that it will in the ",
      "<span style='color:", metric_colours["Profoundly Changed Daily Life in Next 3-5 Years"], "'>Next 3-5 Years</span>"),
       title = "Impact of AI on Daily Life: Past and Future 3-5 Years",
       caption = "Source: Stanford AI Index - Public Opinion (2025). Measures the percentage of survey respondents in a given country who expressed agreement with the specified statement ")

figure4
ggsave(figure4, filename = "figures/figure4.pdf", device = cairo_pdf,
       width = 6, height = 4, units = "in", bg = "transparent")
```

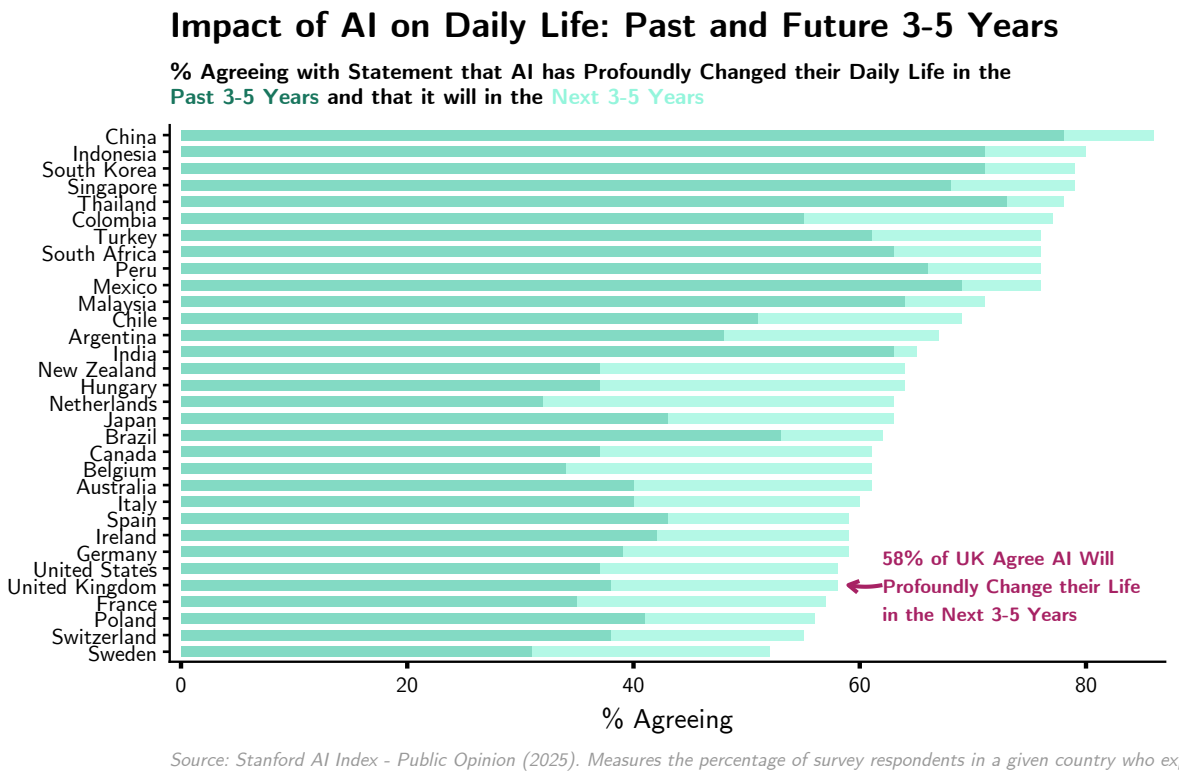


Figure 4 highlights that people in the UK feel like their lives have been, and will continue to be, less impacted by AI than other countries.This is in stark contrast to China and other Asian countries where the public reports very high levels of change due to AI in the past 3-5 years (over 70%) and with even more people believing it will in the future (in China over 80% of those surveyed). Interestingly, the US and other European nations appear to be in a similar position to the UK, with a much lower proportion of people reporting a profound change in their daily lives due to AI so far. In all these countries a higher number of people (typically 20% more) believe that AI will cause profound change in daily lives in the next 3-5 years .

0.5 Figure 5: Excited or Nervous? Lollipop Chart of Public Emotion on AI

As well as reported and expected impact of AI, I was also interested in public emotion about AI, so I analysed the opinion index on the percentage of people that feel excited or nervous about products and services that use AI in each country. Unlike the previous Figure, there was a clear split between countries where people wer more excited than nervous, and vice versa. To visualise this clearly I decided to use a lollipop chart where points correspond to the specific emotion (excited or nervous) and the distance between the two represented by a

line connection.

```
#### Figure 5: Public Opinion Excited vs Nervous about AI
opinion_emotion <- df %>% filter(Source_Category == "Public Opinion") %>%
  filter(Metric %in% c("% Agreeing With Statement: Products And Services Using Artificial Intelligence Make Me Nervous",
                       "% Agreeing With Statement: Products And Services Using Artificial Intelligence Make Me Excited")) %>%
  filter(Year == 2025) %>%
  mutate(Statement = str_trim(str_replace(Metric, "^% Agreeing With Statement:\\s*", ""))) %>% #replace following staement
  group_by(Country) %>%
  mutate(
    excited = Value[Statement == "Products And Services Using Artificial Intelligence Make Me Excited"], #make new columns
    nervous = Value[Statement == "Products And Services Using Artificial Intelligence Make Me Nervous"],
    group = excited > nervous, #group countries by which one is bigger
    order_val = pmin(excited, nervous) #then by values within that group
  ) %>%
  ungroup() %>%
  arrange(group, order_val) %>%
  mutate(Country = factor(Country, levels = unique(Country)))



figure5 <- ggplot(opinion_emotion, aes(x = Country, y = Value, color = Statement)) +
  geom_line(aes(group = Country), color = "grey70", linewidth = 1) +
  geom_point(size = 2) + #so point is on top
  coord_flip()+
  annotate("text",
           x = "United Kingdom", y = 21,
           label = "In the UK AI makes \n38% feel Excited vs\n64% Nervous",
           hjust = 0, size = 2.5, color = "#a72264", fontface = "bold") +
  annotate("curve",
           x = "United Kingdom", xend = "United Kingdom", y = 34, yend = 37,
           arrow = arrow(length = unit(0.15, "cm")),
           curvature = 0.1, color = "#a72264", linewidth = 0.7) +
  scale_y_continuous(limits = c(20, 85), breaks = seq(20, 80, 10),
                     labels = scales::label_percent(scale = 1),expand = c(0, 0))+
  scale_color_manual(values = c("#c83131","#5e72c5")) + #specify colours
  guides(color = "none") +
  theme_ellen() +
  theme(strip.text = element_text(size = 10),
        panel.grid.major.x = element_line(color = "grey80", linewidth = 0.2), # major vertical lines
        panel.grid.minor.x = element_line(color = "grey90", linewidth = 0.1),
        plot.subtitle = element_markdown(size = 8, face = "bold", colour = "black")) +
  labs(
```

```r
    subtitle = paste0( #repeat with colours indicating the statement
      "% Agreeing with Statement that Products and Services using AI make them ",
      "<span style='color:", '#c83131', ";'><b>Excited</b></span>",
      " or ",
      "<span style='color:", '#5e72c5', ";'><b>Nervous</b></span>"),
    title = "Emotional Response to Using AI",
    caption = "Source: Stanford AI Index – Public Opinion (2025). Measures National Average of Survey 'Yes' Responses to Statements",
    x = NULL, y = "% Agreeing"
    )

figure5
ggsave(figure5, filename = "figures/figure5.pdf", device = cairo_pdf,
       width = 6, height = 4, units = "in", bg = "transparent")
```

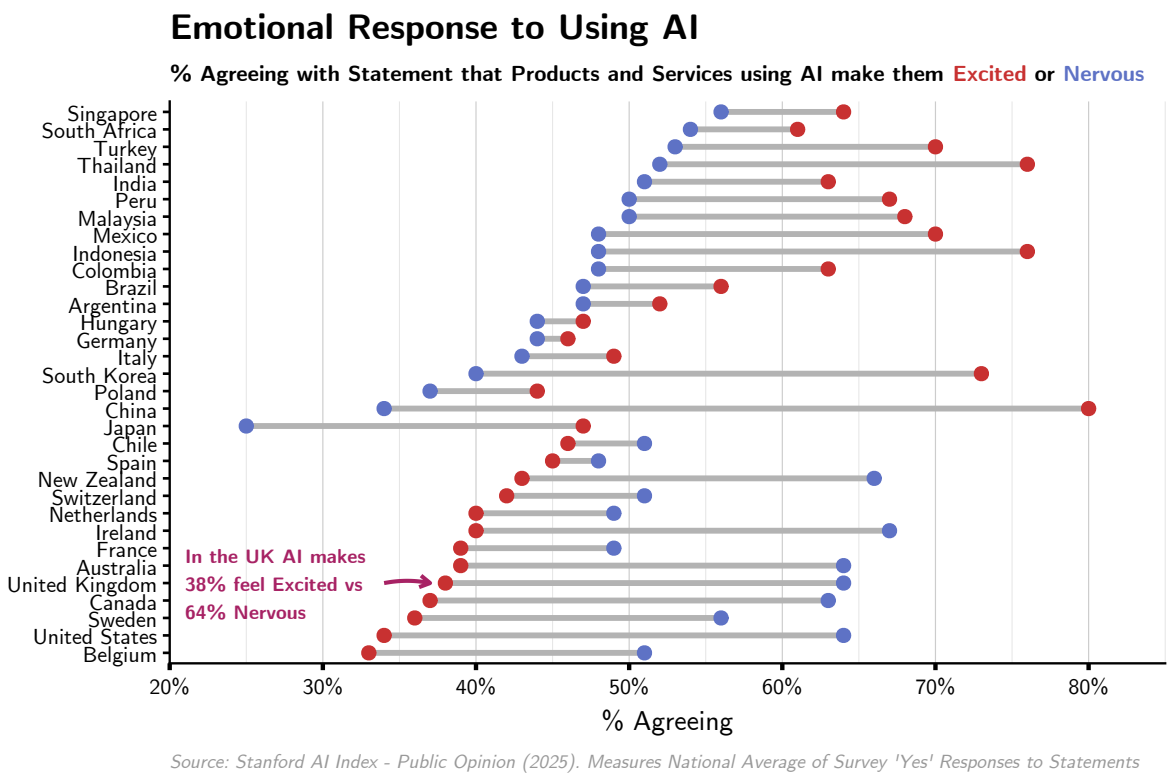


Figure 5 shows a large cohort of countries where more people agreeing with the statement they feel excited about AI product than nervous, whilst another cohort where people feel more nervous than excited (some potentially feeling both). Countries on the top half of the chart fall in the first group, comprising mainly Asian and South American countries, whilst the latter group is primarily European and North American countries, including the UK.
In the UK 64% agree that they feel nervous and 38% feel excited, indicating that generally the UK public is apprehensive about the prospect of AI. The difference between the UK and China is remarkable, with >40% difference in people feeling excited and a >30% difference

in people feeling nervous. There does not appear to be a clear link between economic and R&D capacity (Figure 1) and subsequent public sentiment about AI, indicating other factors likely contribute to emotions regarding AI.

0.6 Figure 6: Map of Global Responsible AI Index Score

Finally, I wanted to look into how the UK performs on metrics reporting ethical use of AI particularly how responsible the UK practices and governance is on ensuring human rights are maintained. Initially I was intrigued whether there are any interesting global patterns in the countries that are more concerned at pursuing AI in a responsible manner, and the performance of the UK in particular. Using the GIRAI Index Score for each country in 2024, I visualised this score on a world map, pairing the colour to the index score to show patterns in country location more clearly.

```r
#### Figure 6: GIRAI Global Responsible AI index score Map
resp_AI <- df %>% #extract index scores
  filter(Source_Category == "Responsible AI") %>%
  filter(Metric == "The Global Index on Responsible AI Score") %>%
  pivot_wider(names_from = Metric, values_from = Value) %>%
  mutate(Country = fct_reorder(Country, `The Global Index on Responsible AI Score`))


resp_AI_index <- resp_AI %>%
  select(ISO3, `The Global Index on Responsible AI Score`) #select just two columns necessary

world_map <- read_sf("/Users/ellen/Documents/GitHub/DataViz_2026/tutorials/Week 5 - EB/ne_110m_admin_0_countries/ne_110m_admin_0_countries.shp")
world_map <- world_map |> filter(ISO_A3 != "ATA") #from tutorial map downloat

combined <- world_map |>
  left_join(resp_AI_index, by = c("ISO_A3" = "ISO3")) #combine the two


figure6 <- ggplot() +
  geom_sf(data = combined,
          aes(fill = `The Global Index on Responsible AI Score`),
          linewidth = 0.25) +
  coord_sf(crs = st_crs("ESRI:54030")) +  # Robinson
  scale_fill_viridis_c(option = "plasma") + #colour
  labs(fill = "Global Index of Responsible AI") + #scale title
  theme_ellen() +
  theme(legend.position = "bottom",
        axis.line.x = element_blank(), #remove all axes details
        axis.line.y = element_blank(),
        axis.ticks.x = element_blank(),
        axis.ticks.y = element_blank(),
```

```
        axis.text = element_blank(),
        axis.title = element_blank(),
        legend.background = element_blank()) +
  labs(title = "2024 Global Index on Responsible AI Score",
       subtitle = "Measures the overall composite performance score (0-100) of a given country across all 185
dimensions and pillars of responsible AI practices and governance.",
       caption = "Source: Global Index on Responsible AI (GIRAI 2024).
Contributing to the score are Dimension on Human Rights and AI, Responsible AI Capacities and Governance,
Government Actions and Frameworks and Non-state actor initiaives on Responsible AI"
       )

figure6
ggsave(figure6, filename = "figures/figure6.pdf", device = cairo_pdf,
       width = 6, height = 4, units = "in", bg = "transparent")
```

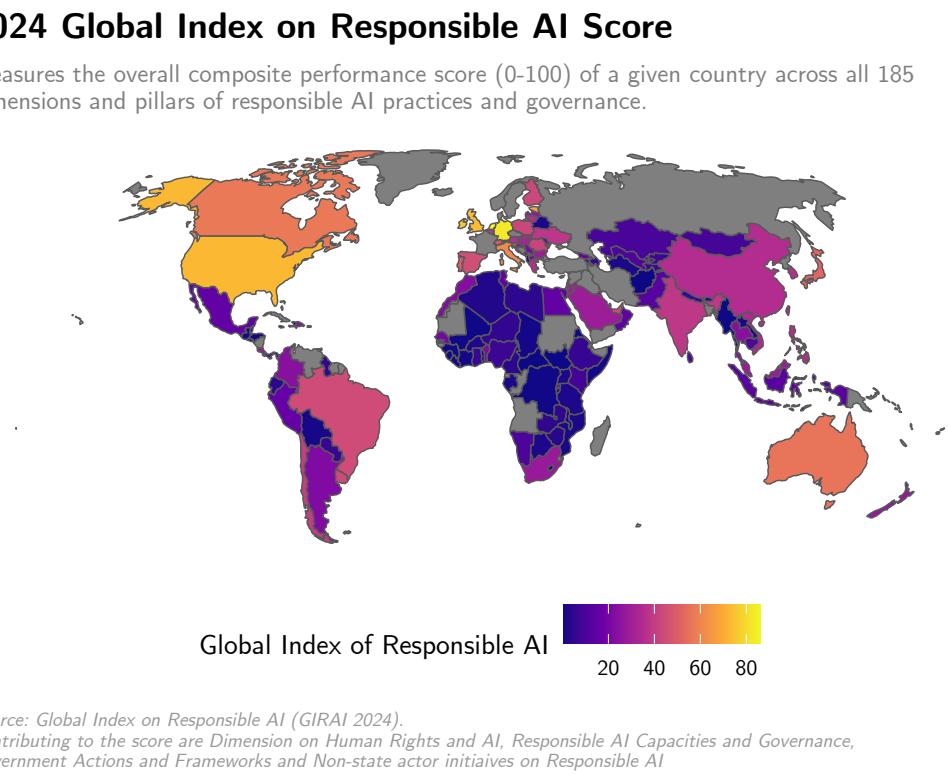


Figure 6 maps countries using the Global Index of Responsible AI, showing that Western nations including the UK are the most responsible in term of AI practice. The UK appears to be at the top end of this score index, on par with the US but behind Germany and the Netherlands. Other countries like Canada, Japan, Australia follow, with China and India scoring around 40 points. Lower levels of responsible practices and governance are seen in South East Asia, South Central America and Africa. This illustrates that the UK is

performing well on Responsible AI metrics with room for improvement.

0.7 Figure 7: Scatter plot of Human Right Score and Trust in AI to not discriminate

Finally, I wanted to explore the relationship between a country's Human Rights with regard to AI and public trust in AI being non-discriminatory. Plotting each country by their GIRAI Dimension score on Human Rights visualises the extent to which they takes steps to protect, promote, and respect human rights implicated by AI such as privacy, equality, and freedom; and the public opinion section of Stanford's AI index as the percentage of individuals that say they trust AI to not discriminate or bias towards any groups.

```r
#### Figure 7: Human Rights and AI Score and Trust in AI to not discriminate/bias
bias <- df %>% filter(Source_Category == "Public Opinion") %>%
  filter(Year == 2024) %>%
  filter(Metric == "% Agreeing With Statement: I Trust Artificial Intelligence To Not Discriminate Or Show Bias Towards Any Group Of People")

rights <-  df %>% filter(Source_Category == "Responsible AI") %>%
  filter(Metric == "DIMENSION SCORES: Human Rights and AI")

comb <- inner_join( #join both metrics
  bias %>% select(Country, bias_score = Value), #match them together by country
  rights %>% select(Country, rights_index = Value),
  by = "Country") %>%
    mutate(Highlight = Country == uk) #highlight


figure7 <- ggplot(comb, aes(x = bias_score, y = rights_index, color = Highlight)) +
  geom_smooth(method = "lm", se = FALSE, linetype = "twodash", color = "grey90") + #back layer line
  geom_point(size = 2) + #add points
  geom_text_repel(aes(label = Country), size = 3, family = 'CMU Sans Serif', show.legend = FALSE) +  # add country labels slightly above points
  scale_y_continuous(limits = c(0, 85), expand = c(0,0))+
  scale_x_continuous(limits = c(35, 85), breaks = seq(30, 80, 10), labels = scales::label_percent(scale = 1)) +
  scale_color_manual(values = c("#5e72c5", "#a72264"))+
  guides(color = "none") +
  theme_ellen() +
  theme(
    panel.grid.major.x = element_line(color = "grey90", linewidth = 0.2),  # major vertical lines
    panel.grid.major.y = element_line(color = "grey90", linewidth = 0.2)
  ) +
```

```r
  labs(title = "Human Rights AI Score vs Public Trust in AI to Not Discriminate",
       subtitle = "Responsible AI score on Human Rights and AI vs % Agreeing with Statement \n 'I Trust Artificial Intelligence To Not Discriminate Or Show Bias Towards Any Group Of People'",
       caption = "Source: GIRAI Index 2024/ Stanford AI Index - Public Opinion (2024).
  Dimension Score represents Country take steps to protect promote and respect human rights implicated by AI.
  Agreement % Measures the percentage of survey respondents in a given country who expressed agreement with the specified statement",
       x = "% Agreeing",
       y = "Human Rights and AI Score"
       )

figure7
ggsave(figure7, filename = "figures/figure7.pdf", device = cairo_pdf,
       width = 6, height = 4, units = "in", bg = "transparent")
```

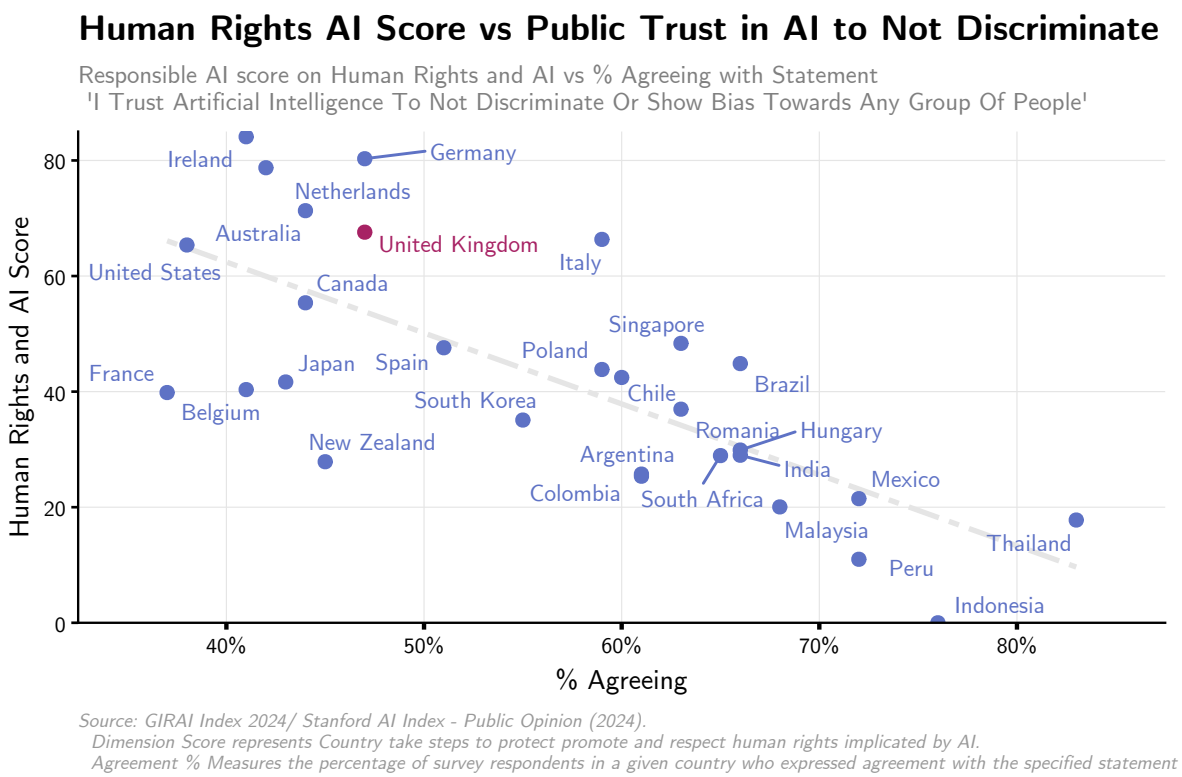


Figure 7 shows that there is a surprising negative relationship between a country's Human Rights score and Trust in AI not to discriminate, i.e. countries with a level of trust tend to rate lower on the human right score. The UK has a moderately high Human Rights and AI score (>60) yet a less then half of the public surveyed said that they trusted AI to not discriminate or show bias. This may indicate that countries in which greater efforts are made

to set up frameworks resulting in the public becoming more aware of the risk of AI being discriminatory. Alternatively it may also be that countries with low levels of public trust in AI put in greater Human Rights Frameworks. A deeper investigation would be necessary to further understand this trend.

Final Figure

Final Figure was compiled on separate txt file and uploaded as a final pdf.

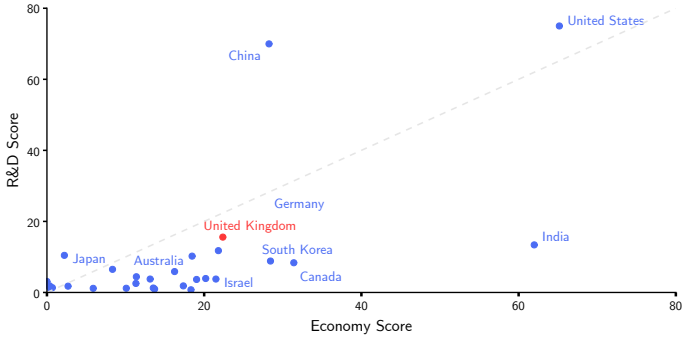
Where is the UK in the AI race?

Ellen Beattie Source: Global Artificial Intelligence Indicator Database (GAID) 1998-2025, sourced from the Harvard Dataverse

Economic Vibrancy

AI Vibrancy: Economy vs Research and Development by Country

Economy and R&D pillar scores for a given country on a 0-100 scale measure the normalised aggregate performance of AI vibrancy metrics

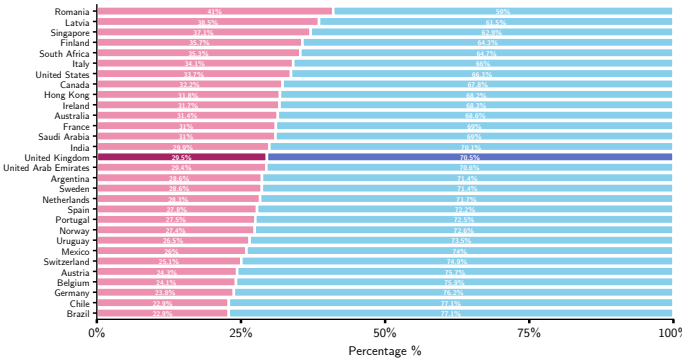


Source: Stanford AI Index - Global AI Vibrancy Tool (2021).
Economic score components include hiring intensity, newly funded companies, investment, skill penetration.
R&D score components include the number of AI journal/conference/public repository publications and citations and AI-related patents

When it comes to AI vibrancy in the Economy (capacity to exploit AI) and Research and Development (innovation capacity), the UK is one of the leaders of the chasing pack below the US, China and India.

2024 AI Talent Representation by Gender

Percentage of AI Specialists Identifying as Female versus Male



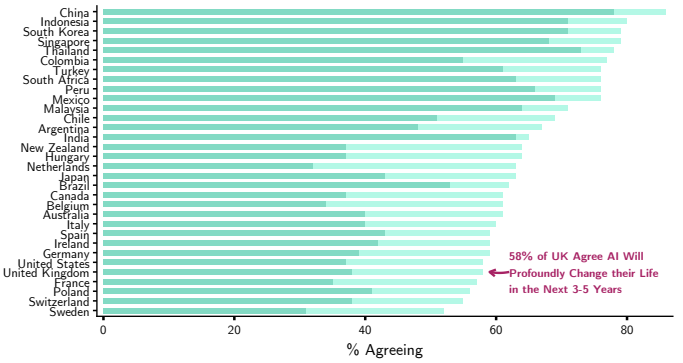
Source: Stanford AI Index - Economy,
Measures the share of the total AI workforce in a given country that identifies as female, indicating gender diversity within th

Women are under-represented in the AI workforce with only 29.5% of the UK AI specialists being female, which is broadly typical of a global issue of gender inequality in AI talent

Public Opinion

Impact of AI on Daily Life: Past and Future 3-5 Years

% Agreeing with Statement that AI has Profoundly Changed their Daily Life in the Past 3-5 Years and that it will in the Next 3-5 Years

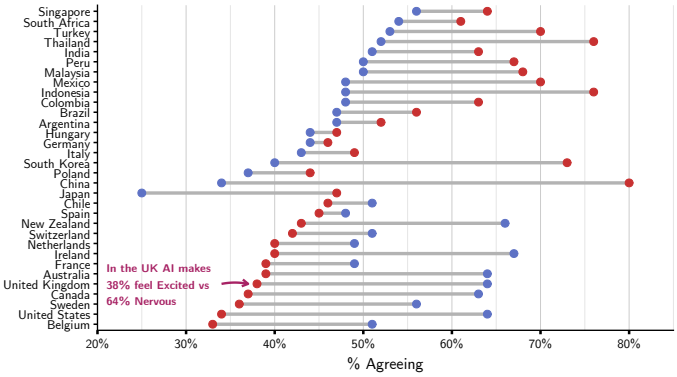


Source: Stanford AI Index - Public Opinion (2025). Measures the percentage of survey respondents in a given country who ex

Fewer people in the UK have felt that their life has been or will be drastically changed by AI compared with most other countries. China, Indonesia and South Korea lead as having the greatest percentage of people reporting and anticipating profound change to their daily lives because of AI.

Emotional Response to Using AI

% Agreeing with Statement that Products and Services using AI make them Excited or Nervous



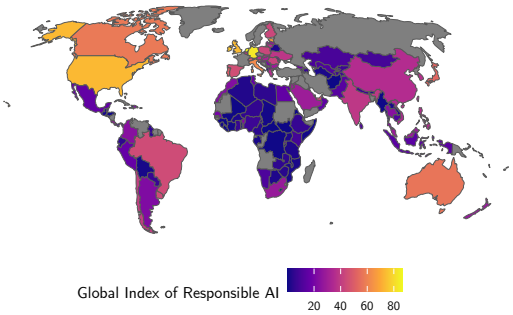
Source: Stanford AI Index - Public Opinion (2025). Measures National Average of Survey 'Yes' Responses to Statements

A substantially higher proportion of the UK public feel more nervous than excited about AI use, a trend seen across other European and North American nations. In other countries, especially those in Asia and South America, more people feel excited and less nervous about AI use.

Ethics and Responsible AI practices

2024 Global Index on Responsible AI Score

Measures the overall composite performance score (0-100) of a given country across all 185 dimensions and pillars of responsible AI practices and governance.

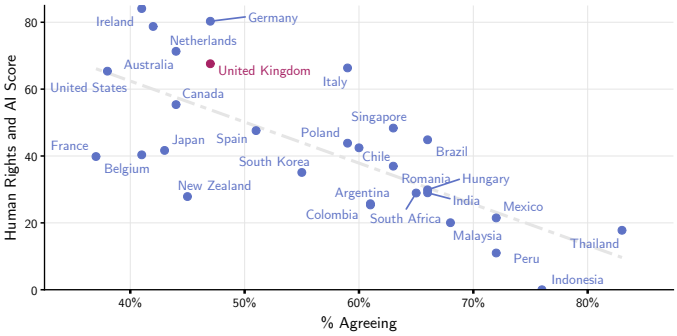


Source: Global Index on Responsible AI (GIRAI 2024).
Contributing to the score are Dimension on Human Rights and AI, Responsible AI Capacities and Governance, Government Actions and Frameworks and Non-state actor initiaives on Responsible AI

Western nations are leading globally as the most responsible in terms of AI practises and governance, with lower scores across South America, Africa and Asia. The UK falls just behind Germany and the Netherlands indicating they are enacting responsible AI initiatives, frameworks and policy to a reasonable extent.

Human Rights AI Score vs Public Trust in AI to Not Discriminate

Responsible AI score on Human Rights and AI vs % Agreeing with Statement 'I Trust Artificial Intelligence To Not Discriminate Or Show Bias Towards Any Group Of People'



Source: GIRAI Index 2024/ Stanford AI Index - Public Opinion (2024).
Dimension Score represents Country take steps to protect promote and respect human rights implicated by AI.
Agreement % Measures the percentage of survey respondents in a given country who expressed agreement with the specified statement

The UK has a moderately high Human Rights and AI score (>60) yet a less then half of the public surveyed said that they trusted AI to not discriminate or show bias. This is consistent with a general negative relationship between a higher public trust in AI to be non-discriminatory, and a lower human rights and AI score.